

Summary:

Around 24 years of experience in Automotive product development predominantly in Electronics and embedded software. Expertise in Quality delivery, collaborate with technology partners and Institutes globally, diplomatically handle conflicts, mentor and appraise team.

Present employment:

Employer : **Mahindra & Mahindra Limited, MRV Chennai**
Designation : **Head - Advanced Engineering and Technology**
Duration : **Sep 2nd 2024 to till date**

Responsibilities:

- Leading dynamically experienced team of Program manager, Mechanical, Electronics and Firmware team.
- Drive technology roadmap for Center of Excellence that includes NVH, Body Engineering, Transmission, Hydraulics, Engine and Advanced Engineering, with Product planning team
- Deliver End-to-End solution for projects like Electric powertrain (Battery, Motor, OBC, DC-DC, PDU, cooling system), Autonomous, technology like Power shuttle, hitch control, Joystick, Loader applications, Telematics, etc.
- Collaborate for International operations for K2 OJA, NYJ products, support for customer care, advanced engineering production setup at Zaheerabad, Nagpur, etc

Previous employment 1:

Employer : **Ipetronik, Bangalore (www.ipetronik.com)**
Designation : **Vice President - Automotive software**
Duration : **Jan 3rd 2024 to Aug 30th 2024**

Role:

Leading R&D in India with 60+ engineers in Software development, Testing team of Manual and Automation, Project Leads, Reporting to Global R&D head in Germany.

Responsibilities:

- Develop skillset for Software development in India to handle customer projects for measurement devices
- Roadmap of projects for India team in cooperation with Global competency heads in locations at Baden-Baden, Dusseldorf and Bergkirchen at Germany

Previous employment 2:

Employer : **Tractors and Farm Equipment Ltd. (TAFE), Chennai**
Designation : **Assistant Chief Engineer**
Duration : **Feb 3rd 2014 to Nov 21st 2023**

Role: Leading embedded software engineering team,

- Design and develop ECUs like Hitch, Power shuttle Transmission, TFT Instrument cluster, EMS along with technology partners
- Basic Software for microcontroller – MCAL, ECAL in manual C code
- Develop and maintain reusable bootloader, UDS on CAN (ISO 14229), SAE J1939
- Adhere to V-model and functional safety (ISO 26262 / ISO 25119). Familiar with FMEA.
- Traceability between Requirements, Design and Testing using PTC ALM
- Conduct technical review including Walkthrough, Inspection and Phase review
- Implement End of Line(EOL) diagnostics tool for vehicle assembly plant

Responsibilities:

- Lead embedded systems development, create roadmap, set goals and drive for success
- Provide technical insights to guide the team
- Build, manage and mentor high performing team including rewards and skill development
- Implement best practices in design and development. Also encourage for continuous improvement
- Project Management for externally awarded projects
- Plan for department budget and provide estimate for project budget
- Establish and maintain cordial relationship with external design partners for development of Telematics, EV motors and Battery Management System

Previous employment 3:

Employer : **Danfoss Industries Pvt. Ltd, Chennai**
Designation : **Project Manager for New Product Development**
Duration : **Apr 10th, 2013 to Jan 10th 2014**

Role: Project Manager for Refrigeration control unit development in matrix organization

Responsibilities:

- Responsible for new product development based on Business case until handing over to line for manufacturing and business owner for sales ramp-up
- Ensure “Product Development Process” is adhered
- Time schedule preparation for new product development milestone deliverable
- Milestone status to Project committee chaired by Sr. VP and members consisting of global head of R&D, Sales, marketing, quality, Supply chain, Finance

Previous employment 4:

Employer : **Jasmin-Infotech Pvt. Ltd, Chennai**
Designation : **Project Leader - Automotive domain**
Duration : **Jan 19th, 2011 to March 29th, 2013**

Role:

- Universelle Handy Vorbereitung(UHV) for Volkswagen. Developed software to interface parrot CK5050+ module to interface mobile phones with instrument cluster and head unit using BAP over CAN. This facilitates hands-free communication, voice command & display.
- Developed CAN communication for Panasonic Headunit

Previous employment 5:

Employer : **WABCO India Limited, Chennai**
Designation : **Senior Software Engineer**
Duration : **May 9th, 2007 to 22nd Oct 2009**

Role: Lead ECAS team in Chennai and coordinate with Hannover, Germany team

Project Description: Electronically Controlled Air Suspension (ECAS) is to maintain the Chassis level of the vehicle irrespective of road conditions and load over it in order to provide smooth and safe driving for Trucks and Buses

- As a technical software architect design and review application software for Electronically Controlled Air Suspension (ECAS)
- Designed and deployed complete application software State Machine (SM)
- Review application software components such for Bus Kneeling, Traction Control Algorithm, Overload Detection, Turnover Protection, Air Consumption and Leakage, etc. are developed as per system requirements
- Developed critical software device driver for Fujitsu MB91F467B microcontroller - Active Height sensor which is used for calculating the position of chassis level.

Previous employment 6:

Employer : **Visteon , Chennai**
Designation : **Software Design Engineer**
Duration : **Sep 27th, 2004 to 3rd May 2007**

Role: Lead Instrument cluster project (FORD) in Chennai and coordinate with North America team

- Technically review and ensure application software architecture for Instrument Cluster as per Visteon standards
- Design, Implement and test software for Instrument Cluster States Machine
- Design, Implement and test application software for TELLTALE and chimes
- Review of Network communication modules using CAN protocol

Previous employment 7:

Employer : **Thomson Multimedia India Pvt. Ltd,**
Designation : **Senior Executive Software,**
Duration : **Oct 9th, 2000 to Sep 24th,2004**

Role: Develop software for Television in Philips microcontroller

- Analyze the requirements, develop software, integrate with Hardware and test it according to requirements on Philips Ultimate One Chip (UOC), SANYO UOC for Indian market.
- Addition of features like Alternate channel, Zoom, Wakeup Timer, video mute, Calculator, 5 Band Graphic Equalizer, Audio and Video modes, Personal Preference setting, Message display, Phone number storage, Program scheduler, Alarm, Global Time, 200 programs, 6 Key Front panel, Wide, Parental Lock.

Technical Skills:

Languages : Embedded C,
Microcontroller (16, 32-bit) : **Fujitsu MB91F467B, Freescale S12X**
Operating System : **Vector OSEK, FREERTOS, DI Kernel**
Comm. Protocol : **SPI, CAN, higher layer protocol – UDS & SAE J1919**
Version Control : **ClearCase, PTC ALM**
Quality standards : **ASPICE 3, MISRA C**
Design tools : **Compiler, debugger, TESSY, QA C, Candela studio,**

Fair knowledge on ORCAD schematic and PCB design tool

Achievements:

- ✓ Established new technical team for ECU development at TAFE
- ✓ Conducted Voice of Customer(VOC) workshop in Italy with department leads from R&D, Product Management, Sales, Supply chain at Danfoss
- ✓ As a fresher made project build setup using Keil uVision for microcontroller at Thomson

Articles published:

- Published article on "Quality Management for Embedded Software" in PMI.org in 2012
<http://www.projectmanagement.com/articles/283968/Quality-Management-for-Embedded-Software>
- Published article on "Project Scope: Managing Complexity & Uncertainty" in PMI.org in 2018
<https://www.projectmanagement.com/articles/433462/Project-Scope--Managing-Complexity-and-Uncertainty>

EDUCATIONAL QUALIFICATION:

Master of Business Administration from University of Madras in the year 2011
Project Management Professional (PMP) certified from PMI by year 2011 (PMP 1458879)
Bachelor of Electronics engineering in 2000, Government college of Engineering, Salem

Personal details:

Date of Birth : **2nd Jul 1979**
Marital status : **Married**